

Operating System (ISE3137)

Jump Together, Fly Farther!



인하대학교 국제학부



Week 10 Lab

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System logging

Processes and the operating system kernel need to be able to record a log of events that happen. These logs can be useful for auditing the system and troubleshooting problems. By convention, the **/var/log** directory is where these logs are persistently stored.

A standard logging system based on the Syslog protocol is built into Red Hat Enterprise Linux. Many programs use this system to record events and organize them into log files. In Red Hat Enterprise Linux 7, syslog messages are handled by two services, **systemd-journald** and **rsyslog**.

The **systemd-journald** daemon provides an improved log management service that collects messages from the kernel, the early stages of the boot process, standard output and error of daemons as they start up and run, and syslog. It writes these messages to a structured journal of events that, by default, does not persist between reboots. This allows syslog messages and events which are missed by syslog to be collected in one central database. The syslog messages are also forwarded by **systemd-journald** to **rsyslog** for further processing.

The **rsyslog** service then sorts the syslog messages by type (or facility) and priority, and writes them to persistent files in the **/var/log** directory.

The **/var/log** directory holds various system- and service-specific log files maintained by **rsyslog**:

ANALYZING AND STORING LOGS

Overview of system log files

Log file	Purpose
<code>/var/log/messages</code>	Most syslog messages are logged here. The exceptions are messages related to authentication and email processing, that periodically run jobs, and those which are purely debugging-related.
<code>/var/log/secure</code>	The log file for security and authentication-related messages and errors.
<code>/var/log/maillog</code>	The log file with mail server-related messages.
<code>/var/log/cron</code>	The log file related to periodically executed tasks.
<code>/var/log/boot.log</code>	Messages related to system startup are logged here.

What is tar?

Archiving and compressing files are useful when creating backups and transferring data across a network. One of the oldest and most common commands for creating and working with backup archives is the **tar** command.

With **tar**, users can gather large sets of files into a single file (archive). The archive can be compressed using **gzip**, **bzip2**, or **xz** compression.

The **tar** command can list the contents of archives or extract their files to the current system. Examples of how to use the **tar** command are included in this section.

Operate the tar command

To use the **tar** command, one of the three following actions is required:

- **c** (create an archive)
- **t** (list the contents of an archive)
- **x** (extract an archive)

Commonly used options are:

- **f *file name*** (file name of the archive to operate on)
- **v** (verbosity; useful to see which files get added to or extracted from the archive)

ARCHIVING AND COPYING FILES

In the following example, an archive named **archive.tar** is created with the contents of file1, file2, and file3 in the user's home directory.

```
[user@host ~]# tar cf archive.tar file1 file2 file3
[user@host ~]# ls archive.tar
archive.tar
```



Note

When archiving files by absolute path names, the leading / of the path is removed from the file name by default. This helps avoid mistakes which could cause important files to be overwritten. Files are normally extracted relative to the current working directory of the tar command.

ARCHIVING AND COPYING FILES

Create the tar archive **/root/etc.tar** with the **/etc** directory as content as user root:

```
[root@host ~]# tar cf /root/etc.tar /etc
tar: Removing leading '/' from member names
[root@host ~]#
```



Important

While **tar** stores ownership and permissions of the files, there are other attributes that are not stored in the **tar** archive by default, such as the SELinux context and ACLs. To store those extended attributes in the **tar** archive, the **--xattrs** option is required when creating an archive.

List contents of a tar archive

To list the content of an archive, the **t** and **f** options, accompanied by the archive to operate, are required.

List the content of the archive **/root/etc.tar**:

```
[root@host ~]# tar tf /root/etc.tar
etc/
etc/fstab
etc/crypttab
etc/mtab
...
```

Extract an archive created with tar

A tar archive should normally be extracted in an empty directory to ensure it does not overwrite any existing files. If files are extracted by root, **tar** attempts to preserve the original user and group ownership of the files. If a regular user extracts files using **tar**, the extracted files are owned by that user.

Extract the archive **/root/etc.tar** to the **/root/etcbackup** directory:

```
[root@host ~]# mkdir /root/etcbackup
[root@host ~]# cd /root/etcbackup
[root@host etcbackup]# tar xf /root/etc.tar
```

By default, when files get extracted from an archive, the umask is subtracted from the permissions of archive content. This is a security measure and prevents extracted regular files from receiving execute permissions by default. To preserve the permissions of an archived file, the **p** option is to be used when extracting an archive.

Extract the archive **/root/myscripts.tar** to the **/root/scripts** directory while preserving the permissions of the extracted files:

```
[root@host ~]# mkdir /root/scripts
[root@host ~]# cd /root/scripts
[root@host scripts]# tar xpf /root/myscripts.tar
```


Create a compressed tar archive

There are three different compression methods supported by the **tar** command. The **gzip** compression is the fastest and oldest one, and is most widely available. The **bzip2** compression usually leads to smaller archive files compared to **gzip** and is less widely available than **gzip**, while the **xz** compression method is relatively new, but usually offers the best compression ratio of the methods available.



Note

The effectiveness of any compression algorithm depends on the exact nature of the data being compressed. Data files that are already compressed, such as compressed picture formats or rpm files, usually lead to a low compression ratio.

It is good practice to use a single top-level directory, which can contain other directories and files, to simplify extraction of the files in an organized way.

To create a compressed tar archive, one of the following **tar** options can be specified:

- **z** for **gzip** compression (filename.tar.gz or filename.tgz)
- **j** for **bzip2** compression (filename.tar.bz2)
- **J** for **xz** compression (filename.tar.xz)

ARCHIVING AND COPYING FILES

Create (c option) a gzip-compressed (z option) tar archive **/root/etcbbackup.tar.gz** of the **/etc** directory on serverX:

```
[root@serverX ~]$ tar czf /root/etcbbackup.tar.gz /etc
```

Create (c option) a bzip2-compressed (j option) tar archive **/root/logbackup.tar.bz2** of the **/var/log** directory on serverX:

```
[root@serverX ~]$ tar cjf /root/logbackup.tar.bz2 /var/log
```

Create (c option) a xz-compressed (J option) tar archive **/root/sshconfig.tar.bz2** of the **/etc/ssh** directory on serverX:

```
[root@serverX ~]$ tar cJf /root/sshconfig.tar.xz /etc/ssh
```

Extract a compressed tar archive

The first step when extracting a compressed tar archive is to determine where the archived files should be extracted to, then create and change to the target directory. To successfully extract the archive, it is usually not necessary to use the same compression option used when creating the archive, as the **tar** command will determine which compression was used. It is valid to add the decompression method to the **tar** options as follows:

Extract (x option) the contents of a gzip-compressed (z option) tar archive named **/root/etcbackup.tar.gz** to the directory **/tmp/etcbackup**:

```
[root@serverX ~]$ mkdir /tmp/etcbackup
[root@serverX ~]$ cd /tmp/etcbackup
[root@serverX etcbackup]$ tar xzf /root/etcbackup.tar.gz
```

Extract (x option) the contents of a bzip2-compressed (j option) tar archive named **/root/logbackup.tar.bz2** to the directory **/tmp/logbackup**:

```
[root@serverX ~]$ mkdir /tmp/logbackup
[root@serverX ~]$ cd /tmp/logbackup
[root@serverX logbackup]$ tar xjf /root/logbackup.tar.bz2
```

ARCHIVING AND COPYING FILES

Extract (x option) the contents of a xz-compressed (J option) tar archive named **/root/sshbackup.tar.xz** to the directory **/tmp/sshbackup**:

```
[root@serverX ~]$ mkdir /tmp/sshbackup  
[root@serverX ~]$ cd /tmp/sshbackup  
[root@serverX sshbackup]# tar xJf /root/sshbackup.tar.xz
```



Note

Listing a compressed **tar** archive works in the same way as listing an uncompressed **tar** archive.



Note

Additionally, **gzip**, **bzip2**, and **xz** can be used independently to compress single files. For example, **gzip etc.tar** results in the compressed file **etc.tar.gz**, while **bzip2 abc.tar** results in the compressed file **abc.tar.bz2** and **xz myarchive.tar** results in the compressed file **myarchive.tar.xz**.

The corresponding decompress commands are **gunzip**, **bunzip2**, and **unxz**. For example, **gunzip /tmp/etc.tar.gz** results in the uncompressed tar file **etc.tar**, while **bunzip2 abc.tar.bz2** results in the uncompressed tar file **abc.tar** and **unxz myarchive.tar.xz** results in the uncompressed tar file **myarchive.tar**.

Overview of tar options

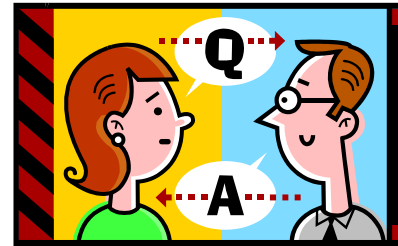
The **tar** command has many options to use. The following table lists some common options and their meanings.

Overview of tar options

Option	Meaning
c	Create a new archive.
x	Extract from an existing archive.
t	List the contents of an archive.
v	Verbose; shows which files get archived or extracted.
f	File name; this option needs to be followed by the file name of the archive to use/create.
p	Preserve the permissions of files and directories when extracting an archive, without subtracting the umask.
z	Use gzip compression (.tar.gz).
j	Use bzip2 compression (.tar.bz2). bzip2 typically achieves a better compression ratio than gzip .
J	Use xz compression (.tar.xz). xz typically achieves a better compression ratio than bzip2 .

Brief break (if on schedule)

Q&A?



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